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Article in *Proceedings of SPIE - The International Society for Optical Engineering* · September 2006

DOI: 10.1117/12.684614

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Performance Comparisons of InGaAs, extended InGaAs, and Short-wave HgCdTe Detectors between 1 μm and 2.5 μm

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ABSTRACT

In this study, three different detectors, regular InGaAs, short-wave infrared extended-InGaAs (exInGaAs) with the bandgap wavelength at 2.6 μm and short-wave HgCdTe (swMCT) with the bandgap wavelength at 2.8 μm are studied. The detectors have active areas of 3 mm or 1 mm diameter with all the detectors capable of being cooled from room temperatures to -85°C with 4-stage thermo-electric coolers. Two of the detectors have field-of-view limiting, cold shrouds attached. From room temperatures to their coldest operating temperatures, the detectors are compared for their temperature-dependent shunt resistances, absolute spectral power responsivities, and noise performances at the output of the photocurrent meter. The photodiode current measuring circuit is analyzed to determine the effect of the shunt resistance for the output offset voltage, the noise and drift amplification, the uncertainty of the current-to-voltage conversion, and the linear operation. The temperature dependences of the shunt resistances are described by Arrhenius plots, and the spectral power responsivities are determined against a pyroelectric detector standard with constant responsivity versus wavelength. We determine that the shunt resistances of regular InGaAs photodiodes can increase to 5 G Ω when cooled to -20°C demonstrating Si-like performance. The shunt resistances of the 1 mm diameter extended InGaAs and short-wave MCT photodiodes were both measured to be about 11 M Ω at diode temperatures of -70°C . Further increase in the shunt resistances would be possible with decreasing diode temperatures. The noise voltage at the output of the photocurrent-to-voltage converter is measured for the respective detectors to determine the noise-equivalent power.

Keywords: infrared detectors, InGaAs, extended InGaAs, MCT, NEP, noise, responsivity, shunt resistance, temperature dependence

1. INTRODUCTION

Although the needs to obtain spectral data from the near-infrared (NIR) and short-wave infrared (SWIR) wavelength regions in remote sensing, astronomical observation, spectroradiometry and spectrophotometry have been increasing^{1,2}, many NIR and SWIR spectrophotometers and spectroradiometers still utilize photoconductive PbS or PbSe detectors or cryogenically-cooled InSb detectors. These detectors are limited in their ability to measure low-level signals due to the low shunt resistances of the detector materials. Furthermore, the cryogenic cooling required for the InSb limits the types of applications with its maintenance needs. The traditionally used PbS and PbSe detectors can be operated in modes which can lead to nonlinear measurements. Since many of the applications have dominant uncertainties which are limited by the signal-to-noise of the measurements, lowering the detection limits by choosing low-noise detectors can lead to reduced uncertainties on the data products.

The widespread use of InGaAs diodes in the telecommunication industry has lead to increased understanding of this material. It is well known that the band-gap wavelength of the material can be tuned by changing the material composition $\text{In}_x\text{Ga}_{1-x}\text{As}$ as grown on InP, such that the higher concentration of indium in the material will lead to a shift of the band-gap wavelength to longer wavelengths³. However, the increase in the indium concentration is also accompanied by increasing strain and material defects in the $\text{In}_x\text{Ga}_{1-x}\text{As}$ layer due to the lattice mismatch which leads to poorer spatial uniformity and lower shunt resistances. The lower shunt resistances can be countered by decreasing the operating temperature of the photodiode for optimum signal-to-noise of the system. In a similar way by changing the

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material composition, the band-gap wavelength is tuned to shorter wavelengths in the short-wave MCT as compared to the traditional use at the 12 μm thermal infrared wavelength region⁴. Both types of detectors can be designed to overcome the thermal background limited performance of InSb by optimizing the band-gap wavelength by material composition. Although in the past each of these detectors have been measured separately, all three detectors have not been compared in a single study of their temperature-dependent shunt resistances and noise-equivalent spectral power responsivities.

To determine the optimal design and operating conditions for near-infrared and short-wave infrared detectors, three different infrared detector types are studied. The detectors have active areas of 3 mm or 1 mm diameter constructed in a hermetically sealed metal packages with sapphire windows with all the detectors capable of being cooled from room temperatures to -85°C with 4-stage thermo-electric cooling. The shunt resistances of the detectors were measured as a function of temperature which were fit with a single activation energy from their Arrhenius plots. The spectral power responsivities and the noise-equivalent power responsivities were measured by comparison to a characterized pyroelectric detector.

2. EXPERIMENTAL DESIGN

All the detectors were purchased from the same vendor with the same detector package for integration into our measurement setup. The detectors were mounted into a water-cooled holder so that the heat generated from the TE cooler could be dissipated. The detector leads were placed into a specialized socket so that the detectors could be easily swapped, and the diode leads were connected to 50 Ω , coaxial wire for the measurement of the photocurrent and shunt resistances. The diode temperatures were stabilized using a commercial TE cooler controller by setting the target thermister resistances attached to the diode.

2.1 Experimental setup for the measurements of shunt resistance

In order to measure the shunt resistance as a function of temperature for each photodiode as in Fig. 1, a light-tight box of anodized aluminum was constructed to house the photodiode apparatus. Once sealed from outside radiation (thus minimizing extraneous photocurrents due to background noise), the photodiode was thermoelectrically cooled. Using the

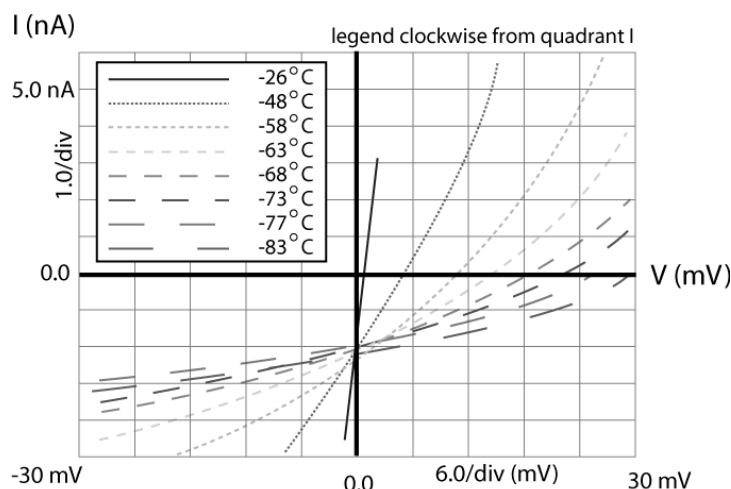


Fig. 1. The current-voltage curves measured for the extended InGaAs diode showing the change in shunt resistances as a function of diode temperature. The curves are shifted from the origin due to the stray incident radiation on the diodes.

Hewlett Packard** 4145A Semiconductor Parameter Analyzer, a voltage sweep was performed across a limited range of voltages while measuring the current in the photodiode. The current was measured over a range of voltages. The shunt resistance for each temperature is taken as the slope of the resulting diode curve as it passes through the y-axis as shown in Fig. 1. These sweeps were conducted for each detector over a range of temperatures from as high as room

temperature to as low as -85 °C. The upper limit of our ability to measure shunt resistances with low uncertainties is < 5 GΩ due to the instrument resolution limits.

2.2 Temperature-dependent shunt resistances

A summary of the temperature-dependent shunt resistances of all the detectors studied can be seen in Fig. 2. The temperature dependences of the shunt resistances of the diodes can be grouped into two groups according to the band-gap energy. All the temperature dependences of the detectors can be well fit by a single slope which corresponds to a single activation energy with the temperature dependent resistances fitted by

$$\Omega(T) = \Omega_o \exp\left(\frac{E_a}{k_B T}\right), \quad (1)$$

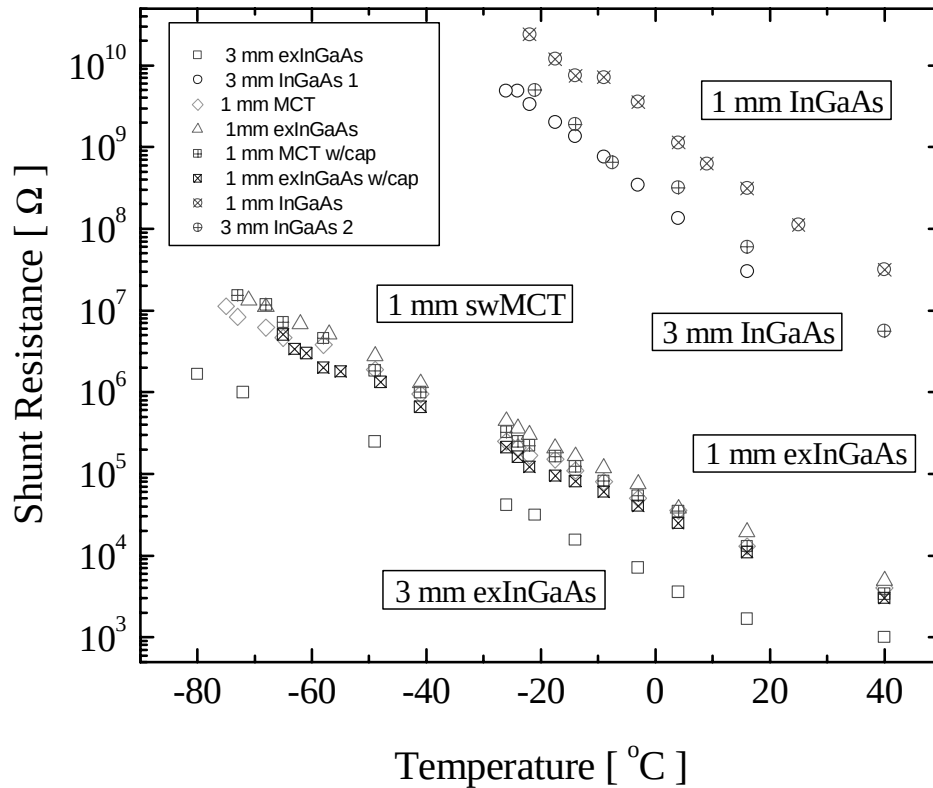


Fig. 2. Temperature dependent shunt resistances of the detectors studied. The regular InGaAs shows high shunt resistances even at room temperature which increases with decreasing temperature. The larger diameter diodes show lower shunt resistances proportional to the active area.

where T is the temperature, k_B is the Boltzmann constant, and E_a is the fitted activation energy. The 1 mm and the 3 mm diameter InGaAs diode start at much higher shunt resistances at room temperature which increase quickly with decreasing temperature. Although the active areas of the two sizes of the InGaAs diodes are different, the slopes seen in Fig. 2 are similar with comparable activation energies. The offset differences of the shunt resistances for the different

diameters of the same material is due to the nature of the defects limiting the shunt resistances which scale with the active area, and thus larger diameter detectors are expected to have lower shunt resistances. The temperature dependent shunt resistances of the short-wave infrared materials are also grouped near the bottom of the figure. Both the 1 mm diameter extended InGaAs and the 1 mm short-wave MCT diodes show similar increase with decreasing temperature. The slope or the activation energy of the short-wave infrared material shows roughly a factor of two difference in the activation energy. The shunt resistances both of the exInGaAs and the swMCT continue to increase with decreasing temperature indicating that at even lower detector temperatures with operation at liquid nitrogen temperatures, even higher shunt resistances are achievable. A summary of the results is shown in Table 1.

Table 1. The number of detectors studied in this work along with the detector parameters. The shunt resistances were measured as described above. The activation energies are obtained from the slope of the curves in Fig. 2. Although lower diode temperatures were achievable for the InGaAs diodes, the higher shunt resistances $> 5 \text{ G}\Omega$ could not be measured with low uncertainties.

Detector number	Detector material	Band-gap wavelength at room temperature [μm]	Detector diameter [mm]	Lowest measured diode temperature [$^{\circ}\text{C}$]	Shunt resistance at lowest temperature [Ω]	Highest measured diode temperature [$^{\circ}\text{C}$]	Shunt resistance at highest temperature [Ω]	Activation Energy [eV]
1	Regular InGaAs	1.7	3.0	-21	5.00×10^9	40	5.60×10^6	0.324
2	Regular InGaAs	1.7	3.0	-26	4.90×10^9	40	2.78×10^6	0.305
3	Regular InGaAs	1.7	1.0	-22	2.40×10^{10}	40	3.15×10^7	0.318
4	Extended InGaAs	2.6	3.0	-80	1.67×10^6	40	1.00×10^3	0.149
5	Extended InGaAs	2.6	1.0	-71	1.33×10^7	40	4.93×10^3	0.169
6	Extended InGaAs with cap	2.6	1.0	-73	1.53×10^7	40	3.40×10^3	0.174
7	Short-wave MCT	2.8	1.0	-75	1.13×10^7	40	4.00×10^3	0.158
8	Short-wave MCT with cap	2.8	1.0	-65	5.06×10^6	40	3.00×10^3	0.167

2.3 Spectral Responsivities

The spectral power responsivities and the signal-to-noise of each of the detectors measured to determine the noise-equivalent power. The spectral power responsivities were measured using a tungsten ribbon lamp source and a Cary 14 prism-grating double monochromator for spectral selection as shown in Fig. 3. The tungsten filament lamp was imaged 1:1 onto the 2 mm wide by 10 mm high entrance slits of the monochromator. The resulting bandwidth of the monochromator ranges from 7 nm to 5 nm depending on the wavelength. The output from the monochromator was directly imaged onto the diodes using a spherical mirror. The photocurrents from the diodes were converted to voltages using a current-to-voltage transimpedance amplifier. The voltages were directly read using a digital voltmeter or fed into a lock-in amplifier using a chopper wheel at the entrance of the monochromator so that comparisons of direct and modulated measurements could be performed. The spectral scans were performed at 50 nm intervals for the various detectors from Table 1.

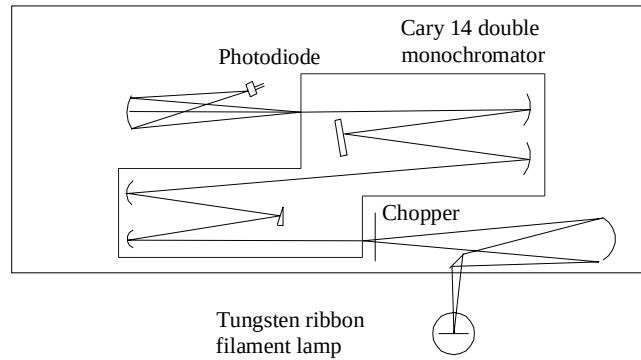


Fig. 3. Experimental setup for the spectral power responsivity measurements for the determination of noise-equivalent power responsivity. The tungsten ribbon lamp was used as a source with the spectral selection by the modified Cary 14 prism-grating double monochromator. The output slit of the monochromator was imaged onto the diodes. Some of the measurements utilized the chopper wheel for modulation of the radiation.

The absolute power responsivity was obtained by using a spectrally flat pyroelectric detector coated with gold-black coating. The pyroelectric detector was calibrated using a Si-trap detector at 442 nm and then the spectral responsivity was calculated to longer wavelengths using the measurement of the spectral reflectance of the pyroelectric detector⁵. If the optical power incident on the detector is known, then the noise-equivalent power (NEP) can be determined using

$$\text{NEP} = \frac{P}{S/N} = \frac{N}{S/P} = \frac{N}{R}, \quad (2)$$

where S is the detector output signal, P is the incident radiant power, R is the detector responsivity, and N is the detector output noise. At each wavelength where the power responsivity is measured, at least 10 measurements are performed to obtain the standard deviation of the measurements. The output noise, N , of the diodes are measured in the dark for DC measurements or with a nominal signal for AC measurements.

Spectral power responsivity

Spectral power responsivities of the regular InGaAs and the extended-InGaAs detectors were measured to test the spectral coverage at high shunt resistances (low detector temperatures) and to determine the NEP. The swMCT detectors could not be measured to their upper cut-off wavelengths because the pre-disperser of the monochromator, used for the spectral responsivity test, dominated the cut-off at 2.6 μm . As an example, the spectral responsivity results obtained on a 3 mm diameter extended-InGaAs test detector are shown in Fig. 4. In order to perform percent level signal-to-noise ratios at the output of the pyroelectric radiometer, a “quasi” detector substitution procedure was introduced. This procedure resulted in the relative spectral power responsivity of the test detector. Both the 3 mm diameter test detector and the 5 mm diameter pyroelectric detector were overfilled by the incident monochromatic radiation and the signal ratios were determined at 50 nm increments from consecutive spectral scans. The relative spectral responsivity was converted into spectral radiant power responsivity using an absolute tie point. The tie point was made against the same pyroelectric radiometer at 1000 nm where the throughput (and the signal) of the monochromator is high. During the tie point transfer, a strict substitution was made. Both the test and the standard detectors measured the same flux (radiant power) by positioning an aperture in front of the pyroelectric detector that has the same area as that of the test detector. The test-detector to standard-detector signal ratio multiplied by the responsivity of the standard-detector resulted in the spectral power responsivity at 1000 nm.

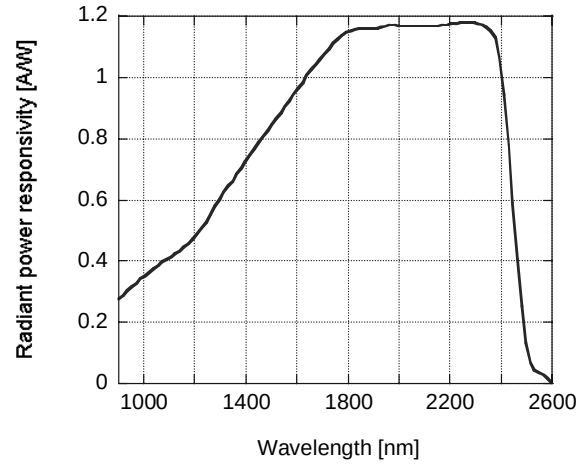


Fig. 4. Spectral power responsivity of a 3 mm diameter ext-InGaAs photodiode.

High (3 s) lock-in time constants were used during all measurements to further increase signal-to-noise ratios at the output of the pyroelectric radiometer. The optical radiation was chopped with a frequency of 13.5 Hz. The 10 % ($k=2$) expanded measurement uncertainty was dominated by the flux non-equivalence on the two detectors and the low signal-to-noise ratios at the output of the pyroelectric radiometer around 2200 nm (where the pre-disperser had a very low throughput).

2.4 Noise measurements.

The output total noise voltage was measured versus signal-gain selections for the 3 mm (3 M Ω shunt resistance) extended-InGaAs photodiode current-meter. The temperature of the detector was controlled to -85 °C using a 4-stage TE cooler. The noise equivalent current (NEC) is equal to the measured total noise-voltage divided by the current-to-voltage gain (the value of the feedback resistor). The total noise-voltage was measured by a lock-in amplifier attached to the output of the photocurrent meter. The chopping frequency was 7.5 Hz and the integrating time constant of the lock-in amplifier was 1 s. The signal (irradiance) source was a 314 °C blackbody radiator located about 1 m away from the detector. The noise equivalent power (NEP) was calculated as the ratio of the NEC to the detector (peak) power responsivity of 1.2 A/W (at 2 μ m). The NEC and NEP results are shown on the two Y axes of Fig. 6. The measured data

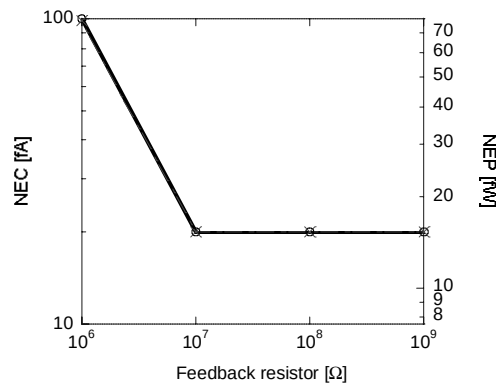


Fig. 5. NEP (right-Y) and NEC (left-Y) of a 3 M Ω shunt resistance ext-InGaAs photodiode current meter at a lock-in integrating time constant of 1 s.

points are connected with straight lines for better illustration. The graph shows that the signal-to-noise ratio could not be improved at the 10^8 V/A and 10^9 V/A signal-gains relative to the 10^7 V/A selection where the shunt resistance is roughly equal to the feedback resistance. From the 10^7 V/A signal-gain (to higher signal-gain selections) the constant $3\text{M}\Omega$ shunt resistance dominates the resistor noise (parallel connection of R_S and R). The 10^8 V/A and 10^9 V/A signal-gains can be utilized only with higher than $3\text{M}\Omega$ shunt resistance detectors when the NEP can be further decreased.

3. DISCUSSION

3.1 Impact of shunt resistance on photocurrent measurements

In order to describe the effects of detector and feedback impedances and noise sources in a photodiode current meter, the equivalent circuit has to be discussed. As it is shown in Fig. 6, a photodiode P can be substituted by an ideal current source I_P and parallel connected shunt resistance R_S and junction capacitance C_J . The photocurrent I_P from photodiode P is converted into a voltage V at the output of the operational amplifier (OA). R is the feedback resistance and C is the feedback capacitance of the OA.

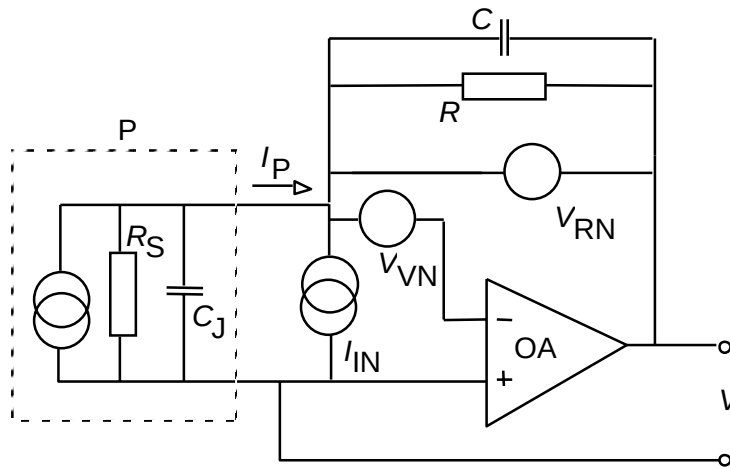


Fig. 6. Equivalent circuit of a photodiode current meter. The physical package of the photodiode is denoted by the dotted line.

The signal gain (current-to-voltage conversion) can be described with the following equations:

$$\frac{V}{I_P} = R \frac{1}{1 + j\omega RC} = R \frac{1}{1 + G^{-1}} \quad (3)$$

where G is the loop gain. G must be large at the signal (e.g. the chopping) frequency to keep the uncertainty of the current-to-voltage gain (R) low. The loop gain, which is the product of the OA open-loop gain A and the feedback attenuation β , is

$$G = A\beta \quad (4)$$

where $\beta = R_S/(R_S + R)$ at low frequency (DC) measurements. High loop gain can be obtained if β is close to unity (its maximum). To achieve this condition, the shunt resistance must be high at the signal frequency.

The closed-loop voltage gain, that determines the amplification of the input noise, offset, and drift for the output of the current-to-voltage converter, is the reciprocal of β :

$$A_V = \frac{R_S + R}{R_S} \quad (5)$$

Again, A_V can be kept small (close to unity) if the shunt resistance is selected to a high value (relative to the feedback resistance). Similarly to the signal-gain above, both G and A_V are frequency dependent⁶.

The input impedance of the above photocurrent meter can be calculated as the parallel connection of the down-transformed feedback resistor and the up-transformed feedback capacitor of the OA:

$$(Z_I)^{-1} = \left(\frac{R}{A}\right)^{-1} + \left(\frac{1}{j\omega AC}\right)^{-1} \quad (6)$$

Linear signal-gain (response) can be performed if the short-circuit photocurrent of the photodiode is measured. In this case, the photodiode shunt impedance must be much larger than the input impedance of the current meter:

$$Z_I \ll Z_p \quad (7)$$

3.2 The Impact of the Temperature-dependent shunt resistances

The shunt resistances of detectors in the short-wave infrared wavelength region, can be increased easily by lowering the temperature of the detectors from room temperature from $\sim \text{k}\Omega$ to $\sim \text{M}\Omega$. Such increase in the shunt resistance is critical if low uncertainty photocurrent measurements are to be performed at preamplifier gains equal to the reciprocal of the shunt resistances of gain 10^6 V/A or 10^7 V/A . We have shown for the regular InGaAs photodiodes that the shunt resistances equal to that of Si diodes are possible by cooling the diodes to -20°C or lower. These cooled detectors can then be operated with gains of 10^{10} V/A with low uncertainties due to low noise and nonlinearity effects. If the short-wave infrared detectors continue to follow a semi-logarithmic relationship with temperature then it is expected that these detectors would also achieve shunt resistances $> 10 \text{ G}\Omega$ ⁷. The temperature extrapolation is shown in Fig. 7 and plotted in kelvin to illustrate the low temperature relationship. Similar relationship is expected for the short-wave MCT detectors.

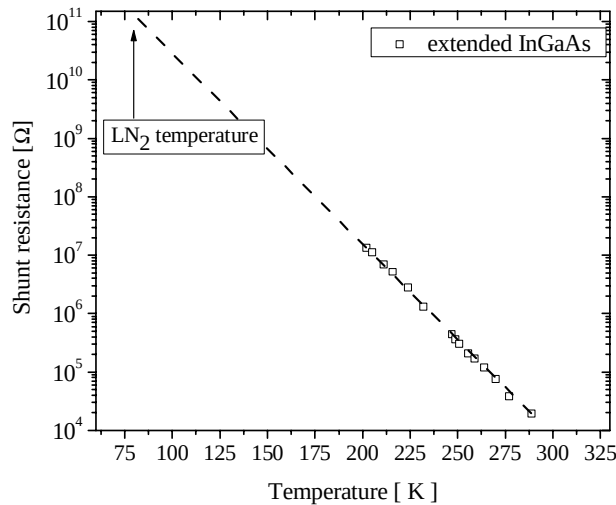


Fig. 7. The temperature dependent shunt resistances of extended InGaAs with a line extrapolated to liquid nitrogen temperature of 77 K demonstrating possible shunt resistances $> 10 \text{ G}\Omega$.

3.3 Areal Dependence of the Shunt Resistance

Another relationship that we can observe from Table 1 is the areal dependence of the shunt resistance as plotted in Fig. 8. The shunt resistances of the InGaAs and the extended InGaAs detectors are plotted as a function of the area for the lowest measured diode temperatures. The short-wave MCT is not plotted since only 1 mm diameter detectors were examined. Although the diodes are comprised of different materials, the physical defects limiting the shunt resistance are uniformly distributed across the diode surface leading to a linear dependence on the diode area. A linear dependence of the shunt resistance with area would fall on the straight lines. This linear dependence is observed for Si diodes with high shunt resistances which are expected to have low defect densities. From the observed linear dependence in Fig. 8,

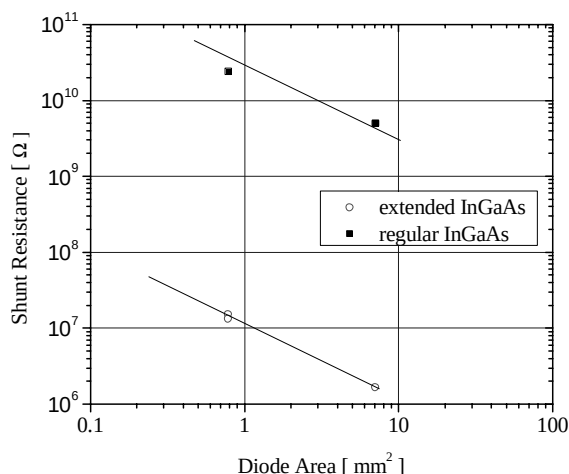


Fig. 8. The dependence of the shunt resistance on the area of the InGaAs and the extended InGaAs diodes. The straight lines indicate the linear dependence between shunt resistance and diode surface area.

the spatial uniformity of the original wafer material is expected to be high. Although larger diodes are expected to have lower shunt resistances, the shunt resistances can be increase back to that of smaller diodes by decreasing the operating temperature.

4. CONCLUSIONS

The temperature dependence of shunt resistances of InGaAs, extended InGaAs and short-wave HgCdTe diodes have been measured. The temperature dependences have been fit using an Arrhenius plot and two different activation energies have been found. The importance of the shunt resistances on the photocurrent measurements have been illustrated using an equivalent circuit model. Extrapolation of the shunt resistances to liquid nitrogen temperatures would imply that optical measurements at gains of 10^{10} V/A would be possible in the short-wave infrared wavelength region from 1.0 μm to 3.0 μm enabling Si-type performance. We determine that the shunt resistances of the regular InGaAs increase to $> 5 \text{ G}\Omega$ with cooling of the diode to -20°C demonstrating Si-type performance. The shunt resistances of the 1 mm diameter extended InGaAs (exInGaAs) and short-wave HgCdTe (swMCT) were both measured to be $\sim 11 \text{ M}\Omega$ at diode temperatures of -70°C . Further increase in the shunt resistances would be possible with decreasing diode temperatures. These measurements demonstrate the clearly better diode characteristics as compared to traditional PbS or InSb detectors used in these wavelength regions. These detectors hold much promise for lowering the near infrared and short-wave infrared measurement uncertainties in commercial spectrophotometers, remote sensing and astronomical applications.

ACKNOWLEDGMENTS

Matt C. Dopkiss was supported by the National Science Foundation under Agreement No. PHY-0453430. Any opinions, findings, and conclusions or recommendations expressed are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

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